

The demand for convenient and reliable switching systems is increasing across homes, offices, and industrial environments. Conventional methods, such as wired switches and infrared (IR) remotes, have clear limitations: wired systems require complex installation and lack mobility, while IR control is effective only over short distances and requires a direct line of sight. Radio frequency (RF) technology overcomes these constraints by enabling longer-range, robust wireless communication, making it well-suited for automation.

operating range is 30-50 metres indoors and up to 100 metres in open outdoor areas, depending on antenna quality and installation conditions.

Circuit and working

Transmitter unit. The transmitter is a compact, battery-powered circuit equipped with a push-button that sends control commands. It is built around the TX2H-433-64 RF module, operating at 433.92MHz, and powered by a regulated +5V DC supply, which can be provided by a rechargeable battery, coin-cell arrangement, or USB adaptor.

approximately 17 centimetres long, which corresponds to a quarter wavelength at this frequency. A telescopic antenna may be used if an extended range is required.

Receiver unit. Fig. 2 shows the receiver unit. It uses the RX2A-433-64 RF module to detect the transmitted signal. The signal is amplified by transistor BC107 (T1) and then supplied to a CD4027 dual J-K flip-flop integrated circuit (IC1A) configured as a bistable latch. Each activation of switch S1 at the transmitter toggles the output state of the flip-flop, thereby switching the load on or off.

Since the flip-flop output cannot directly drive the relay, a relay driver transistor 2N2219 (T2) is used. Transistor T2 supplies sufficient current to energise the relay coil (RL1). When T2 conducts, current flows through the relay coil, activating it and switching the connected AC load. A 5V SPDT (single-pole double-throw) is used to control a 230V AC appliance such as a lamp, fan, or small motor.

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